

LegalEase – Program Code

Code snippets arranged from the provided project documentation.

1. main.py

```
from fastapi import FastAPI
from legalEaseAPI.routes import router

app = FastAPI(title="LegalEase - AI Legal Document Generator")

# Include API routes
app.include_router(router)

# Root endpoint
@app.get("/")
def home():
    return {"message": "Welcome to LegalEase AI Legal Document Generator API"}

# Run FastAPI if executed directly
if __name__ == "__main__":
    import uvicorn
    uvicorn.run(app, host="0.0.0.0", port=8000)
```

2. routes.py

```
from fastapi import APIRouter

from pydantic import BaseModel

from ai_core.gemini_generator import GeminiDocumentGenerator

router = APIRouter()

gemini_generator = GeminiDocumentGenerator()

class DocumentRequest(BaseModel):
    document_type: str
    parties: str
    terms: str
    dates: str

@router.post("/generate")
def generate_legal_document(request: DocumentRequest):
    response = gemini_generator.generate_document(
        request.document_type,
        request.parties,
        request.terms,
        request.dates
    )

    return {"document": response}
```

3. gemini_generator.py

```
def generate_document(self, document_type, parties, terms, dates):
    prompt = (
        f"Generate a comprehensive legal document titled {document_type}\n"
        f"Involved parties: {parties}\n"
        f"Effective Date: {dates}\n"
        f"Terms and Conditions: {terms}\n"
        "Ensure formal legal structure with multiple sections and legal clauses."
    )
```

```
response = self.model.generate_content(prompt)

return response.text
```

4. Gemini API code

```
genai.configure(api_key=os.getenv("GEMINI_API_KEY"))

self.model = genai.GenerativeModel(model_name)

response = self.model.generate_content(prompt)
```

5. frontend/app.py – Page configuration

```
st.set_page_config(page_title="LegalEase", layout="centered")
```

6. frontend/app.py – Logo layout

```
col1, col2, col3 = st.columns([1, 2, 1])

with col2:
    st.image(WEB_LOGO_PATH, use_container_width=True)
```

7. frontend/app.py – Header

```
st.markdown(
    "<h2 style='text-align: center;'>AI Legal Document Generator</h2>",
    unsafe_allow_html=True
)
```

8. frontend/app.py – User input

```
document_type = st.text_input("Document Type")

parties = st.text_area("Parties Involved")

terms = st.text_area("Terms & Conditions (Use semicolons for bullet points)")

dates = st.text_input("Effective Date")
```

9. frontend/app.py – Send request

```
response = requests.post("http://localhost:8000/generate", json={...})
```

10. frontend/app.py – Full request

```
response = requests.post(
    "http://localhost:8000/generate",
    json={
        "document_type": document_type,
        "parties": parties,
        "terms": terms,
        "dates": dates
    }
)
```

11. Sanitize generated text

```
generated_text = sanitize_text(response.json()["document"])
```

12. HTML Preview

```
styled_html = format_html_preview(generated_text)
```

```

st.markdown(
    f"<div style='{...}'>{styled_html}</div>",
    unsafe_allow_html=True
)

```

13. Edit Document

```

edited_text = st.text_area(
    "Edit Document Below:",
    generated_text,
    height=300
)

```

14. Edit with session state

```

if st.session_state.get("show_edit"):

    edited_text = st.text_area(
        "Edit Document Below:",
        generated_text,
        height=300
    )

    st.session_state.generated_text = edited_text

```

15. Download TXT

```

st.download_button(
    "■ Download as .TXT",
    data=generated_text,
    file_name=f"{document_type.replace(' ', '_').lower()}.txt"
)

```

16. Download DOCX

```

st.download_button(
    "■ Download as .DOCX",
    data=format_docx(generated_text, document_type),
    file_name=f"{document_type.replace(' ', '_').lower()}.docx"
)

```

17. Download PDF

```

st.download_button(
    "■ Download as .PDF",
    data=format_pdf(generated_text, document_type),
    file_name=f"{document_type.replace(' ', '_').lower()}.pdf"
)

```

18. main.py – Route inclusion

```

app.include_router(router)

```

19. routes.py – Generate endpoint

```

@router.post("/generate") # Route for generating legal document

```